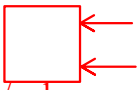
Timer:
Timers use the internal clock cycles generated by the oscillator circuit as measure of a delay.
In the Intel's 8051 versions, these function as follows -
For Every 12 oscillators cycles is measured as period of 1 Machine Cycle and hence this 1 Machine Cycle is measure of the mininum delay that can be produced by timers.

In the "Timer" operation, the timer register is incremented for every machine cycle - that is it counts machine cycles
Hence, the count rate is 1/12 of the oscillator frequency

Simply you can say, Timer is used for counting the internal events (machine cycles).

Counter: Counter is used for counting the external events.



Counters are peripherals which count off external clock cycles/pulses applied onto P3.4 & P3.5 and this gives count of external events.

In the counter function , the external input pin T0 (P3.4) or T1 (P3.5) is sampled during S5P2 of every machine cycle. When the samples show a high in one and a low in the next cycle,the count is incremented.

The new count value appears in the register during S3P1of the cycle following the one in which the transition was detected.
Since it takes 2 machine cycles (24 oscillator periods) to recognize a l-to-O transition, the maximum count rate is 1/24 of the oscillator frequency.

SFRs Associated with Timer/Counter Peripherals: TMOD,TCON,TH0,TL0,TH1,TL1

TMOD is the sfr used for configuring whether peripheral is to used as timer/counter.
Default is Timer. And also mode of operation being used.

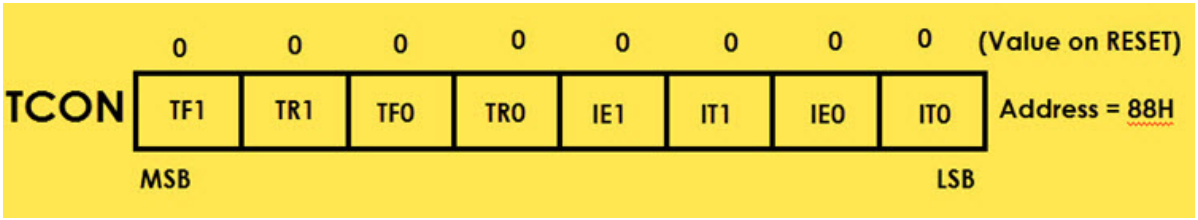
TCON is the sfr used for starting/stoping the timing/counting functionality,checking the overflow flags related to the timers.

2 16-bit TIMERS are available in 8051.

TH0,TL0 are the counting registers of the Timer/Counter 0 peripheral.
TH1,TL1 are the counting registers of the Timer/Counter 1 peripheral.

These timer registers are available in cascaded mode.
TL0 - Timer0 lower byte TH0 - Timer0 higher byte
TL1 - Timer1 lower byte TH1 - Timer1 higher bytes

TCON - it is bit as well as byte addressable register



TF1 Time 1 Overflow flag. Set when the timer rolls from all 1s to 0.
cleared when processor vectors to execute interrupt service routine located at program address 001Bh.

TR1 Timer 1 run control bit.
Set to 1 by program to enable/start the timer
cleared to zero by program to stop/halt timer.

TF0 Timer 0 Overflow flag. Set when the timer rolls from all 1s to 0.
cleared when processor vectors to execute interrupt service routine located at program address 000Bh.

TR0 Timer 0 run control bit.
- Set to 1 by program to enable/start the timer
- cleared to zero by program to stop/halt timer.

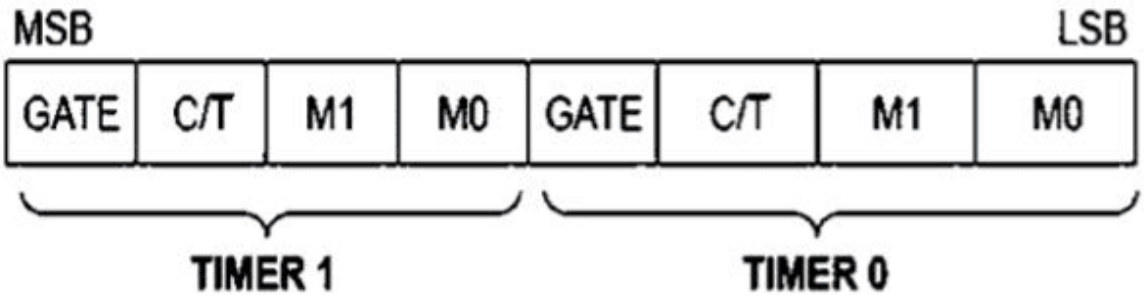
IE1 External interrupt 1 Edge flag.
This bit is set by the processor when there is an interrupt at INT1.
This bit cleared by the processor when there in a jump to ISR of INT1

IT1 External interrupt 1 signal type control bit .
- Set to 1 by program to enable external interrupt 1 to be triggered by a falling edge signal.
- Set to 0 by program to enable external interrupt 1 to be triggered by a low level signal.

IE0 External interrupt 0 Edge flag.
This bit is set by the processor when there is an interrupt at INT0.
This bit cleared by the processor when there in a jump to ISR of INT0

IT0 External interrupt 0 signal type control bit .
- Set to 1 by program to enable external interrupt 0 to be triggered by a falling edge signal.
- Set to 0 by program to enable external interrupt 0 to be triggered by a low level signal.

TMOD - it is byte addressable register



GATE OR gate enable bit which controls RUN/STOP of timer 1/0.
- Set to 1 by program to enable timer to run if bit TR1/0 in TCON register is set and signal on external interrupt INT1/0 pin is high.
- Cleared to 0 by program to enable timer to run if bit TR1/0 in TCON is set.

C/T(bar) Set to 1 by program to make timer 1/0 act as a counter by counting pulses from external input pins 3.5 (T1) or 3.4(T0).
Cleared to 0 by program to make timer act as a timer by counting internal frequency

M1 Timer/ Counter operating mode select bit 1.
- Set / Cleared by program to select mode.

M0 Timer/ Counter operating mode select bit 0.
- Set / Cleared by program to select mode.

M1	M0	Mode	
0	0	0	13-bit timer/counter mode
0	1	1	16-bit timer/counter mode
1	0	2	auto reload mode
1	1	3	split timer mode

M1	M0	Operating mode
0	0	13 bit Timer / counter: THx with TLx as 5 bit prescaler --- 2^13 = 8192/ 256*32=8192
0	1	16 bit Timer / counter: THx and TLx are cascaded --- 256*256 = 65536
1	0	8 bit auto reload Timer / Counter THx holds a value which is to be reloaded into TLx each time it overflows
1	1	(Timer 0) TL0 is an 8-bit Timer / Counter controlled by the standard Timer 0 control bits. TH0 is an 8 bit timer only controlled by Timer1 control bits (Timer 1) Timer / Counter 1 stopped

Time delay calculation:

Tdelay = (2^x - initial value to be loaded from where count is started) * MCT
where x = 13, for mode0
x = 16, for mode1
x = 8, for mode2&mode3

Fosc = 12MHZ,
MCT = 12/Fosc
= 12/12MHZ
= 1usec

for example, 1msec delay is required to create using timer0 mode0.

1msec = (2^13 - initial value)*1usec
1000 usec = (8192 - initial value)*1usec

initial value = 8192 - 1000 = 7192

